Metastatic Risk for Distinct Patterns of Postirradiation Local Recurrence of Posterior Uveal Melanoma

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Objective: The purpose of the study is to compare the prognostic significance of horizontal/marginal versus vertical/diffuse patterns of postirradiation local recurrence of posterior uveal melanoma.

Design: The study design was a nonrandomized, retrospective clinical study. Semiparametric and nonparametric statistical techniques were used.

Participants: Seven hundred sixty-six posterior uveal melanoma patients were studied.

Intervention: Either iodine-125 plaque or helium ion radiation therapy was performed.

Main Outcome Measures: Local tumor recurrence and systemic metastasis were measured.

Results: Local tumor recurrence was detected in 66 (8.6%) of 766 irradiated tumors. The 5-year actuarial rate of local recurrence was 10%. The recurrence pattern was horizontal/marginal in 27 patients (41%) and vertical/diffuse in 39 patients (59%). Systemic metastasis was detected in 5 patients (19%) with horizontal/marginal recurrence and in 19 patients (49%) with vertical/diffuse recurrence. After known metastatic risk factors were controlled, the relative risk for metastasis was 2.2 for horizontal/marginal recurrence and 5.1 for vertical/diffuse recurrence ($P = 0.05$). The actuarial rate of systemic metastasis was 2.9% per year for all patients, 6.3% per year for patients with horizontal/marginal recurrence, and 15.5% per year for patients with vertical/diffuse recurrence.

Conclusions: Postirradiation local recurrence of posterior uveal melanoma is a risk factor for systemic metastasis. Vertical/diffuse recurrences may be associated more strongly with metastatic disease than horizontal/marginal recurrences.

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Postirradiation local recurrence of posterior uveal melanoma is an occasional complication of both episcleral plaque brachytherapy and charged particle teletherapy. Local failure has been reported in 4% to 18% of patients treated with radioactive plaques[1–7] and 2% to 5% of patients treated with charged particle therapy (helium ions or protons).[8,9] Several studies have reported an increased risk for systemic metastasis and melanoma-related mortality in patients with local recurrence.[4,8,10] However, none of these reports distinguished between different morphologic patterns of local melanoma recurrence.

Local recurrences may present predominantly as a thin, horizontal extension along a tumor margin (horizontal/marginal recurrence) or as an area of focally or diffusely increased tumor thickness (vertical or diffuse recurrence).

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